TRI-OCTAGON MATHEMATICAL RESEARCH

Recoverable Transfer and Projection Capacity in the Reciprocal Chiral Exchange Candidate

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Abstract

The reciprocal chiral exchange candidate couples a native state to a periodic carrier and a reversible receiver through two finite distributed profiles. We distinguish the fraction of transferred storage that returns from the amount the fixed registration permits to leave the core. For arbitrary admitted preparations and every finite positive constant receiver coupling, the rank-two projection supplies a conserved, generally indefinite native scalar. Together with the published complex bilinear invariant, it yields a sharper all-time ceiling on signed net native debit. We determine the exact minimum on the full common native invariant level, characterize zero capacity by zero raw drive, and exhibit an active coherent launch for which the old ceiling is strictly unreachable. Algebraic sharpness does not imply dynamical attainment.

Separately, for the specified coherent family $\Omega_0 = (1, e^{i\phi}, 1)$, $0 < \phi < \pi/12$, with nonzero chiral drive and initially empty carrier and receiver, an explicit sufficient finite-coupling regime guarantees temporary receiver concentration, later receiver backflow and net native recovery on one unchanged solution. Receiver share and net recovery are at least 80/81 of the stated intermediate transferred stock. The actual maximum stock on the prescribed window has a positive inverse-square leading scale in receiver coupling, while endpoint stock is bounded by a fourth inverse power. The proof controls propagation and native drift at finite coupling and separately bounds receiver power. High recovered fractions therefore coexist with small absolute transfer and necessary structural ceilings; neither a completed spatial circuit nor physical energy calibration follows.

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1 Introduction

The reciprocal chiral extension of the Tri-Octagon model supplies matching native, carrier and receiver accounts through two finite profiles [1]. Its published same-trajectory theorem then connected initial supply from an empty carrier and receiver to later total native credit on one constant-parameter solution [2, Theorem 3.1]. That sign theorem left two quantitative questions distinct: what fraction of an actually transferred stock returns, and how much native storage can be extracted in the first place?

This paper presents the accepted accessible-transfer and projection-capacity investigations [3, 4]. The structural result applies to the general published preparation class at every fixed finite positive receiver coupling. The fixed real rank-two registration has a one-dimensional kernel. Projection onto that line produces a second conserved native quantity and a necessary extraction ceiling that can be strictly smaller than the previously published one. We prove its exact algebraic content, including feasibility and equality states, and distinguish those states from endpoints reached by the coupled dynamics.

The quantitative result has narrower hypotheses: the specified coherent launch family, empty external preparation and an explicit sufficient receiver-coupling threshold. On a fixed comparison-time window, an estimate for the actual solution proves positive transferred stock, its temporary concentration in the receiver, subsequent net native recovery and exact receiver charging and backflow. The recovered fraction can be close to one while the stock itself is small. The native-credit witness is imported from the published theorem; the receiver-power estimates are separate conclusions.

Section 2 fixes the shared model without restricting its preparation. Section 3 proves the projection-capacity results. Sections 4–5 state and prove the finite transfer certificate. Section 6 compares amounts and ceilings while preserving the original normalization of accessible transfer. All statements concern exchange time. The intended geometric route remains a subsequent correspondence problem.

2 Fixed candidate and inherited foundations

2.1 Registration and finite ports

Write $x = (a, b) \in \mathbb{R}^6$, $\Omega = a + ib \in \mathbb{C}^3$, and $I = |\Omega|^2$. Retain the fixed registration and raw port order of the published candidate [1, Section 1]:

$$\begin{gathered} e_y = (0,1,0), \quad e_z = (0,0,1), \quad n_\pm = (2e_y \pm \sqrt{3}e_z)/\sqrt{7}, \quad A_{\mathrm{reg}} = \lambda_g R_z(\delta_g), \\ K = \kappa_{\mathrm{in}} \begin{pmatrix} n_+^T \\ n_-^T \end{pmatrix} A_{\mathrm{reg}}, \qquad f(x) = K(a \times b), \qquad J(x) = D_x f(x). \end{gathered} \tag{2.1}$$

The gains $\lambda_g, \kappa_{\mathrm{in}} > 0$ and yaw δ_g are fixed. The real matrix K has rank two; J is 2×6. Contact-tangent orientation signs do not insert extra signs into this raw chiral projection.

Let $s \in \mathbb{R}/L_c\mathbb{Z}$, with finite $L_c > 0$ and speed $c > 0$. Contacts $s_\pm = L_c/2 \mp \eta_{\mathrm{geom}}$ have shortest periodic distance $2\eta_{\mathrm{geom}}$. Fix $0 < \epsilon_c < \eta_{\mathrm{geom}}$ and set

$$\begin{gathered} b_\sigma(s) = \epsilon_c^{-1} \left(1 - \frac{|\operatorname{wrap}_{L_c}(s - s_\sigma)|}{\epsilon_c}\right)_+, \qquad \mathcal{B}v = b_+ v_+ + b_- v_-, \\ \mathcal{B}^* p = \begin{pmatrix} \int b_+ p \, \mathrm{d}s \\ \int b_- p \, \mathrm{d}s \end{pmatrix}, \quad m = \frac{2}{3\epsilon_c}, \quad d = \frac{2}{\epsilon_c^3}, \\ \mathcal{B}^*\mathcal{B} = m\mathrm{Id}_2, \qquad \|\mathcal{B}\| = \|\mathcal{B}^*\| = \sqrt{m}, \qquad \|\mathcal{B}'v\|_2 = \sqrt{d}\,|v|. \end{gathered} \tag{2.2}$$

All spatial integrals cover one period. The shortest-distance condition ensures disjointness on the whole circle, including its coordinate seam. Each normalized tent is continuous and belongs to H^1_{per}; its first weak derivative is piecewise constant in L^2, without corner delta terms. The displayed identities follow from the affine halves and disjoint support [1, Section 3]. No second profile derivative is used.

2.2 Equations, stores and solution class

Fix $\alpha_N, \beta_R, \eta_N > 0$ and any finite constant $\kappa = \eta_R > 0$. For independent native, periodic carrier and receiver preparations, retain

$$\boxed{\begin{aligned} \dot{x} &= -\frac{\eta_N}{2\alpha_N} J(x)^T y, \quad p_t + c(r_c)_s = \mathcal{B}\{\eta_N f(x) - \kappa z\}, \\ (r_c)_t + c p_s &= 0, \qquad\qquad \dot{z} = \frac{\kappa}{\beta_R} y, \qquad y = \mathcal{B}^* p. \end{aligned}} \tag{2.3}$$

There are no native F jumps, parameter switches or additional feedback terms. Both couplings occupy the same finite profile regions. The trial stores and total signed powers are

$$\begin{gathered} H_N = \frac{\alpha_N}{2}|x|^2, \quad H_C = \frac{1}{2}\int (p^2 + r_c^2)\,\mathrm{d}s, \quad H_R = \frac{\beta_R}{2}|z|^2, \quad H_{\mathrm{tot}} = H_N + H_C + H_R, \\ P_{NC} = \eta_N f(x)^T y, \qquad P_{CR} = \kappa z^T y, \\ \dot{H}_N = -P_{NC}, \qquad \dot{H}_C = P_{NC} - P_{CR}, \qquad \dot{H}_R = P_{CR}, \qquad \dot{H}_{\mathrm{tot}} = 0. \end{gathered} \tag{2.4}$$

The quadratic H_N is distinct from the native quartic potential. Both powers sum all signed port contributions.

The global solution theorem of the reciprocal-exchange paper [1, Theorem 3.1] gives a unique solution in $C(\mathbb{R}; \mathbb{R}^6 \times {L^2_{\mathrm{per}}}^2 \times \mathbb{R}^2)$, with C^1 native and receiver coordinates, conservation and the integrated balances (2.4). The bounded adjoint makes y continuous even for L^2 carrier preparation. Positive storage weights and conservation bound its norm. For H^1 carrier preparation, additionally

$$(p, r_c) \in C(\mathbb{R}; H^1_{\mathrm{per}})^2 \cap C^1(\mathbb{R}; L^2)^2. \tag{2.5}$$

The projection theorem needs only the general class. The quantitative theorem uses empty external preparation, whose carrier belongs to H^1. Existence is inherited, not inferred from a formal energy calculation.

The previously proved complex bilinear invariant is [1, Theorem 5.1]

$$\begin{gathered} Z_2(\Omega) = \Omega^T \Omega = |a|^2 - |b|^2 + 2i\, a \cdot b, \qquad Z_2(\Omega(t)) = Z_0, \\ I^2 - |Z_2|^2 = 4|a \times b|^2, \qquad I(t) \geq |Z_0|, \qquad C_Z := \frac{\alpha_N}{2}(I_0 - |Z_0|) \geq 0. \end{gathered} \tag{2.6}$$

Here and below a transpose is bilinear; a dagger denotes conjugate transpose.

2.3 The special launch used later

The quantitative result will impose, separately from the general model,

$$\Omega_0 = (1, e^{i\phi}, 1), \quad 0 < \phi < \pi/12, \qquad p(0) = r_c(0) = z(0) = 0. \tag{2.7}$$

All moduli are one and the ordinary phase span is ϕ. With $G = \kappa_{\mathrm{in}} \lambda_g$, the published launch evaluation [2, Section 2.3, (2.12)–(2.13)] gives

$$\begin{gathered} R = |x_0| = \sqrt{3}, \qquad k = \|K\| = G\sqrt{8/7}, \qquad F_{\max} = kR^2/2, \\ f_{0,\pm} = \frac{G \sin\phi}{\sqrt{7}}(-2\sin\delta_g \pm \sqrt{3}), \qquad |f_0|^2 = \frac{G^2 \sin^2\phi}{7}(6 + 8\sin^2\delta_g) > 0. \end{gathered} \tag{2.8}$$

An individual component may vanish, but the vector never does for a fixed yaw. No invariance of this launch family or its initial coherent phase sector is presumed.

3 Projection capacity at arbitrary finite coupling

3.1 Invariant and exact native debit

The published rotation convention expands the fixed matrix as

$$K = \frac{G}{\sqrt{7}} \begin{pmatrix} 2\sin\delta_g & 2\cos\delta_g & \sqrt{3} \\ 2\sin\delta_g & 2\cos\delta_g & -\sqrt{3} \end{pmatrix}, \qquad (3.1)$$
$$e = R_z(-\delta_g)e_x = (\cos\delta_g, -\sin\delta_g, 0), \qquad Ke = 0, \quad \ker K = \mathbb{R}e, \quad e \cdot K^T y = 0.$$

The row sum and difference are independent; equivalently $\det(KK^T) = 48G^4/49 > 0$. Define the real scalar

$$h_e = |e \cdot \Omega|^2, \qquad \mathcal{J}_K = I - 2h_e = \Omega^\dagger(\mathrm{Id} - 2ee^T)\Omega. \qquad (3.2)$$

It may be negative and is distinct from the Jacobian $J(x)$. Changing e to $-e$ changes neither quantity. This kernel vector is unrelated to the notation for carrier energy density.

Theorem 3.1 (Projection invariant and joint ceiling)**.** *For the general preparations and fixed parameters of Section 2, $\mathcal{J}_K$ is conserved. If $j_0 = \mathcal{J}_K(\Omega_0)$, then at every exchange time*

$$H_N(0) - H_N(t) = \alpha_N[h_e(0) - h_e(t)]. \qquad (3.3)$$

Moreover,

$$\begin{aligned} I(t) &\geq \max(|Z_0|, j_0), \\ H_N(0) - H_N(t) &\leq C_{\mathrm{joint}} := \frac{\alpha_N}{2}[I_0 - \max(|Z_0|, j_0)] \\ &= \min\{C_Z, \alpha_N h_e(0)\} \geq 0. \end{aligned} \qquad (3.4)$$

For empty external preparation this yields

$$0 \leq H_C(t) + H_R(t) = H_N(0) - H_N(t) \leq C_{\mathrm{joint}} \qquad (t \geq 0). \qquad (3.5)$$

For arbitrary preloaded preparation the signed debit instead equals

$$H_N(0) - H_N(t) = [H_C(t) + H_R(t)] - [H_C(0) + H_R(0)]. \qquad (3.6)$$

Thus the ceiling does not bound the entire pre-existing external stock.

Proof. Put $w = K^T y$ and $\mu = \eta_N/(2\alpha_N)$. The exact native equation [1, Proposition 2.1] is

$$\dot{a} = \mu w \times b, \qquad \dot{b} = -\mu w \times a, \qquad \dot{\Omega} = -i\mu w \times \Omega. \qquad (3.7)$$

Native coordinates are C^1 and w is continuous in the general solution class, so ordinary finite-dimensional differentiation is valid. Triple products give

$$\dot{I} = 2\mu\{a \cdot (w \times b) - b \cdot (w \times a)\} = -4\mu w \cdot (a \times b). \qquad (3.8)$$

Choose a right-handed real orthonormal basis with third vector e. In this basis $w = (w_1, w_2, 0)$, and

$$\begin{aligned} \dot{h}_e &= 2\mu\{a_3(w_1 b_2 - w_2 b_1) - b_3(w_1 a_2 - w_2 a_1)\} \\ &= -2\mu w \cdot (a \times b), \qquad \dot{\mathcal{J}}_K = \dot{I} - 2\dot{h}_e = 0. \end{aligned} \qquad (3.9)$$

This is a coordinate proof of an invariant relation, not a new registration. Since $I(t) = j_0 + 2h_e(t)$, the native store gives (3.3). Its derivative agrees with $\alpha_N \dot{h}_e = -\eta_N w \cdot (a \times b) = -P_{NC}$.

Nonnegativity of h_e gives $I(t) \geq j_0$; combine it with (2.6). Both terms in the minimum in (3.4) are nonnegative, including at native zero. Conservation gives (3.5) and (3.6). No intensity normalization, carrier differentiation, large-coupling condition or coherent-state hypothesis entered. □

The exact debit identity expresses the existing account; the indefinite scalar is not an additional physical store.

3.2 Feasibility and sharp algebraic minimum

Proposition 3.2 (Full common invariant level)**.** *Fix $Z \in \mathbb{C}$ and $j \in \mathbb{R}$. In the full native space $\mathbb{C}^3$,*

$$\begin{aligned} \{\Omega : Z_2(\Omega) = Z,\ \mathcal{J}_K(\Omega) = j\} \neq \varnothing \quad &\Longleftrightarrow \quad j \geq -|Z|, \\ \min_{\{Z_2 = Z,\ \mathcal{J}_K = j\}} |\Omega|^2 = \max(|Z|, j) \quad &(j \geq -|Z|). \end{aligned} \tag{3.10}$$

For infeasible pairs the set is empty and no minimum is asserted.

Proof. Write $\Omega = he + v_\perp$, where $h \in \mathbb{C}$ and $e^T v_\perp = 0$. Real orthogonal decomposition gives

$$Z_2 = h^2 + v_\perp^T v_\perp, \quad \mathcal{J}_K = \|v_\perp\|^2 - |h|^2, \quad I = |h|^2 + \|v_\perp\|^2 . \tag{3.11}$$

The norm is Hermitian, $\|v_\perp\|^2 = v_\perp^\dagger v_\perp$, whereas the bilinear square can vanish at a nonzero vector. The triangle inequality in a real orthonormal basis of $e^\perp$ gives $|v_\perp^T v_\perp| \leq \|v_\perp\|^2$, hence

$$|Z| \geq |h|^2 - |v_\perp^T v_\perp| \geq |h|^2 - \|v_\perp\|^2 = -j. \tag{3.12}$$

This proves necessity. Every feasible state also satisfies $I \geq |Z|$ and $I \geq j$.

For sufficiency and equality put $\rho = |Z|$, and take real orthonormal u, v in $e^\perp$. If $\rho > 0$, choose a unit complex γ with $\gamma^2 \rho = Z$. If $\rho = 0$, set $\gamma = 1$, without division or an argument of zero.

For $-\rho \leq j \leq \rho$, the state

$$\Omega_* = \gamma \left[\sqrt{\frac{\rho + j}{2}}\, u + \sqrt{\frac{\rho - j}{2}}\, e \right] \tag{3.13}$$

has $I_* = \rho$, $Z_2(\Omega_*) = \gamma^2 \rho = Z$ and $\mathcal{J}_K(\Omega_*) = (\rho + j)/2 - (\rho - j)/2 = j$. For $j \geq \rho$, use

$$\Omega_* = \gamma \left[\sqrt{\frac{j + \rho}{2}}\, u + i \sqrt{\frac{j - \rho}{2}}\, v \right] . \tag{3.14}$$

Here $I_* = j$, $h_e = 0$ and $\mathcal{J}_K = j$, while the bilinear square is $\gamma^2[(j + \rho)/2 - (j - \rho)/2] = Z$. All square-root arguments are nonnegative in their declared ranges. These constructions attain the lower bound.

For $\rho > 0$, $j = -\rho$ gives $\gamma\sqrt{\rho}\, e$, and $j = \rho$ gives $\gamma\sqrt{\rho}\, u$ in both branches. Feasible negative j is covered by (3.13). If $Z = 0$, negative j is infeasible; $j = 0$ has minimum zero, uniquely attained at native zero; and $j > 0$ has minimizer $\sqrt{j/2}(u + iv)$. Native zero has $I = Z_2 = \mathcal{J}_K = h_e = 0$. The zero-invariant level can contain nonzero states, as shown below. No compactness assertion about an unbounded level set is required. □

This is sharpness under the two invariant constraints in the full native space. It is not a minimization restricted to the coherent family or a claim that a coupled trajectory reaches a constructed state.

3.3 Zero drive and a strict active witness

Proposition 3.3 (Zero ceiling and zero drive)**.** *For the fixed real rank-two K and any native state,*

$$C_{\text{joint}}(\Omega_0) = 0 \quad \Longleftrightarrow \quad f(x_0) = 0. \tag{3.15}$$

With empty external preparation, zero drive is stationary, whereas nonzero drive gives initial strict supply.

Proof. The minimum in (3.4) is zero precisely when $C_Z = 0$ or $h_e(0) = 0$. In the first case, $I = |Z_2|$ and (2.6) implies $a \times b = 0$. In the second, $e \cdot a = e \cdot b = 0$, so $a \times b$ is parallel to e, possibly zero. Both give $f = 0$.

Conversely, $f = 0$ means $a \times b \in \mathbb{R}e$. If the cross product is zero, (2.6) and nonnegativity give $I = |Z_2|$, hence $C_Z = 0$. If $a \times b = \lambda e$ with $\lambda \neq 0$, its orthogonality to a and b gives $e \cdot a = e \cdot b = 0$, hence $h_e = 0$. This includes native zero without dividing by intensity or chirality.

When $f(x_0) = 0$ and the external state is empty, the constant full state solves (2.3); uniqueness gives stationarity [1, Section 5]. For $f(x_0) \neq 0$, the empty-launch proposition [1, Proposition 4.1] gives

$$H_N(0) - H_N(t) = \frac{\eta_N^2 m}{2} |f(x_0)|^2 t^2 + o(t^2) > 0 \tag{3.16}$$

for all sufficiently small positive t, at each fixed finite coupling. □

A nonzero zero-drive core with preloaded carrier need not be stationary, because $J^T y$ can be nonzero. Native zero is different: $f(0) = J(0) = 0$ for every external state.

For the admitted launch (2.7), direct evaluation gives

$$\begin{gathered} I_0 = 3, \quad Z_0 = 2 + e^{2i\phi}, \quad |Z_0| = \sqrt{9 - 8\sin^2\phi}, \\ h_e(0) = |\cos\delta_g - \sin\delta_g e^{i\phi}|^2 = 1 - \sin(2\delta_g)\cos\phi, \quad j_0 = 1 + 2\sin(2\delta_g)\cos\phi. \end{gathered} \tag{3.17}$$

Choose the single fixed yaw $\delta_g = \pi/4$ before the trajectory. Because $0 < \cos\phi < 1$ and both compared quantities are positive,

$$\begin{gathered} j_0 = 1 + 2\cos\phi > |Z_0|, \qquad j_0^2 - |Z_0|^2 = 4\cos\phi(1 - \cos\phi) > 0, \\ 0 < C_{\text{joint}} = \alpha_N(1 - \cos\phi) < C_Z = \frac{\alpha_N}{2}(3 - \sqrt{9 - 8\sin^2\phi}), \\ |f_0|^2 = \frac{10G^2\sin^2\phi}{7} > 0. \end{gathered} \tag{3.18}$$

The last identity is the published drive (2.8) at this yaw. Thus the old ceiling is separated by a fixed positive gap from every actual net debit, while the launch remains active. The new ceiling is not asserted attained. This witness is neither a yaw classification nor a preferred physical orientation.

3.4 Why algebraic equality need not be reached

For real unit $u \perp e$, the admitted state $\Omega_0 = u + ie$ satisfies

$$I_0 = 2, \quad Z_0 = 0, \quad h_e(0) = 1, \quad j_0 = 0, \quad f(x_0) = K(u \times e) \neq 0. \tag{3.19}$$

Indeed $u \times e$ is a nonzero vector perpendicular to e, so it is not in $\ker K$. This example lies outside the restricted coherent family and may be paired with empty external preparation. Its common native invariant level contains native zero, the unique intensity minimizer on that level.

Suppose the nonzero native state reached zero at a finite time t_*. Retain the actual carrier and receiver coordinates at that time; they need not vanish. The invariant native-zero subspace supplies an evolution through this same full state, since $f(0) = J(0) = 0$. The global forward/backward uniqueness of the published solution theorem forces the original solution to have zero native coordinate also initially, a contradiction. Equivalently, along the actual solution the continuous $w(t)$ is fixed and (3.7) is a homogeneous linear ODE in Ω; uniqueness in both directions gives the same conclusion. This treats the actual coefficient, not a freely chosen control. Nothing here excludes infinite-time approach or contradicts the accepted coherent-launch results.

4 Quantitative transfer on a prescribed window

Now impose exactly the coherent empty preparation (2.7). Fix the geometry, profiles, speed, registration, positive scales, native coupling and one admitted phase before choosing κ. Different values of κ compare different constant-parameter solutions.

4.1 Constants and times

Alongside $R, k, m, d, F_{\max}$ in (2.2) and (2.8), define

$$\begin{gathered} A = \frac{\eta_N^2 k^2 R^2 m F_{\max}}{4\alpha_N}, \qquad \omega = \sqrt{m/\beta_R}, \quad S = \frac{2\pi}{\omega}, \quad T = \frac{3\pi}{2\omega}, \\ \Lambda = \sqrt{\beta_R}|f_0| > 0, \qquad D = \frac{m}{\omega}|f_0|^2 = \omega\Lambda^2 > 0, \\ B_1 = \frac{4c\sqrt{d}\,|f_0|}{\omega^2}, \quad B_2 = \frac{\sqrt{m}\,AS^3}{3}, \quad \varepsilon_\kappa = \frac{B_1}{\kappa} + \frac{B_2}{\kappa^2}, \quad a_\kappa = \frac{\eta_N^2}{2\kappa^2}. \end{gathered} \tag{4.1}$$

Retain the published native-credit constants without change [2, Section 3]:

$$\begin{gathered} C_1 = \frac{3c\sqrt{md}\,|f_0|^2}{\omega^2}, \qquad C_2 = mAT^3(F_{\max} + |f_0|/3), \\ \kappa_{\mathrm{crit}} = \max\{1, 4C_1/D, \sqrt{4C_2/D}\}. \end{gathered} \tag{4.2}$$

For a fixed proof tolerance $0 < q \le 1/5$, set

$$\kappa_{\mathrm{cert}}(q) = \max\left\{\kappa_{\mathrm{crit}}, 1, \frac{2B_1}{q\Lambda}, \sqrt{\frac{2B_2}{q\Lambda}}\right\}. \tag{4.3}$$

All quantities defining this threshold are finite and independent of κ. The tolerance is an analytical accuracy choice, not a physical coefficient or fitted fraction.

The four distinct landmarks and the actual stock maximum are

$$\begin{gathered} t_{\mathrm{q}} = \frac{\pi}{2\omega\kappa} < t_\pi = \frac{\pi}{\omega\kappa} < t_{\mathrm{b}} = \frac{3\pi}{2\omega\kappa} < t_{2\pi} = \frac{2\pi}{\omega\kappa} = \frac{S}{\kappa}, \\ E_{\mathrm{out}}(t) = H_C(t) + H_R(t) = H_N(0) - H_N(t), \qquad M_\kappa = \max_{0\le t\le t_{2\pi}} E_{\mathrm{out}}(t). \end{gathered} \tag{4.4}$$

Continuity on a compact interval supplies this maximum. Its location is not identified: the times are comparison landmarks, not asserted exact extrema or an exact period.

4.2 The finite certificate

Theorem 4.1 (Temporary receiving and net recovery)**.** *For every fixed finite $\kappa \geq \kappa_{\mathrm{cert}}(q)$, the solution from (2.7) satisfies $\varepsilon_\kappa \leq q\Lambda < \Lambda$ and*

$$\begin{aligned} 0 < a_\kappa(2\Lambda - \varepsilon_\kappa)^2 \leq E_{\mathrm{out}}(t_\pi) \leq M_\kappa \leq a_\kappa(2\Lambda + \varepsilon_\kappa)^2, \\ E_{\mathrm{out}}(t_{2\pi}) \leq a_\kappa \varepsilon_\kappa^2. \end{aligned} \tag{4.5}$$

The separate intermediate projections obey

$$H_C(t_\pi) \leq a_\kappa \varepsilon_\kappa^2, \qquad H_R(t_\pi) \geq a_\kappa(2\Lambda - \varepsilon_\kappa)^2 > 0. \tag{4.6}$$

Both normalization denominators are therefore positive. Define

$$\begin{aligned} &\mathcal{S}_\kappa = \frac{H_R(t_\pi)}{E_{\mathrm{out}}(t_\pi)}, \qquad \mathcal{R}_\kappa = \frac{H_N(t_{2\pi}) - H_N(t_\pi)}{E_{\mathrm{out}}(t_\pi)} = 1 - \frac{E_{\mathrm{out}}(t_{2\pi})}{E_{\mathrm{out}}(t_\pi)}, \\ &r_\kappa = \left(\frac{\varepsilon_\kappa}{2\Lambda - \varepsilon_\kappa}\right)^2. \end{aligned} \tag{4.7}$$

Then

$$\begin{aligned} 1 - r_\kappa \leq \mathcal{S}_\kappa, \mathcal{R}_\kappa \leq 1, \qquad 0 \leq \frac{H_R(t_{2\pi})}{H_R(t_\pi)} \leq r_\kappa, \\ \mathcal{S}_\kappa, \mathcal{R}_\kappa \geq 1 - \left(\frac{q}{2-q}\right)^2 \geq \frac{80}{81}, \qquad \frac{H_R(t_{2\pi})}{H_R(t_\pi)} \leq \left(\frac{q}{2-q}\right)^2 \leq \frac{1}{81}. \end{aligned} \tag{4.8}$$

The absolute recovered amount satisfies

$$H_N(t_{2\pi}) - H_N(t_\pi) \geq \frac{2\eta_N^2}{\kappa^2}\Lambda(\Lambda - \varepsilon_\kappa) > 0. \tag{4.9}$$

The receiver powers have separate finite guarantees:

$$P_{CR}(t_{\mathrm{q}}) \geq \frac{\eta_N^2 D}{2\kappa} > 0, \qquad P_{CR}(t_{\mathrm{b}}) \leq -\frac{\eta_N^2 D}{2\kappa} < 0. \tag{4.10}$$

At the same later time, the unchanged published native-credit theorem [2, Theorem 3.1], using $\kappa \geq \kappa_{\mathrm{crit}}$, gives

$$P_{NC}(t_{\mathrm{b}}) \leq -\frac{\eta_N^2 D}{2\kappa} < 0. \tag{4.11}$$

Initially, P_{NC}, H_R, P_{CR} are strictly positive for all sufficiently small positive time. There is a later receiver-backflow interval about t_{b}, as specified in (5.15). All events occur on this same unchanged solution.

The receiver share and recovery use the actual intermediate external stock as denominator. Retention uses the separately positive intermediate receiver stock. Neither is a fraction of initial native storage, a structural ceiling or gross positive work; no tagged cohort is followed. The recovery is net over the stated interval, not monotonic native recovery throughout it. The proof follows.

5 Finite-error proof of the transfer certificate

5.1 Exact scaling and inherited bounds

For each finite $\kappa > 0$, use the published change of variables [2, Section 4.1]:

$$\begin{gathered}\tau = \kappa t, \quad x_\kappa(\tau) = x(\tau/\kappa), \quad (p, r_c, z)(\tau/\kappa) = \frac{\eta_N}{\kappa}(\widehat{p}, \widehat{r}, \widehat{z})(\tau) = \frac{\eta_N}{\kappa} V_\kappa(\tau), \quad \widehat{y} = \mathcal{B}^* \widehat{p}, \\ (x_\kappa)_\tau = -\frac{\eta_N^2}{2\alpha_N \kappa^2} J(x_\kappa)^T \widehat{y}, \qquad \widehat{p}_\tau + \frac{c}{\kappa}\widehat{r}_s = \mathcal{B}(f(x_\kappa) - \widehat{z}), \\ \widehat{r}_\tau + \frac{c}{\kappa}\widehat{p}_s = 0, \qquad \widehat{z}_\tau = \widehat{y}/\beta_R.\end{gathered} \tag{5.1}$$

The original system still has $c > 0$. Since κ is dimensionless, τ retains time units. Stock and power convert with different factors:

$$E_{\mathrm{out}}(\tau/\kappa) = a_\kappa \, \|V_\kappa(\tau)\|_{CR}^2 \,, \qquad P_{CR}(\tau/\kappa) = \frac{\eta_N^2}{\kappa} \widehat{z}^T \widehat{y}. \tag{5.2}$$

For $W = (q_c, h, v)$ define the weighted norm and finite operator

$$\begin{gathered}\|W\|_{CR}^2 = \|q_c\|_2^2 + \|h\|_2^2 + \beta_R |v|^2, \\ \mathcal{L}_\kappa W = \left(-\frac{c}{\kappa} h_s - \mathcal{B} v, \; -\frac{c}{\kappa}(q_c)_s, \; \frac{1}{\beta_R}\mathcal{B}^* q_c \right), \qquad D(\mathcal{L}_\kappa) = {H^1_{\mathrm{per}}}^2 \times \mathbb{R}^2.\end{gathered} \tag{5.3}$$

Periodic integration and adjointness give

$$\begin{gathered}\langle W, \mathcal{L}_\kappa W \rangle_{CR} = -\frac{c}{\kappa} \int (q_c h_s + h(q_c)_s) \, \mathrm{d}s - \langle q_c, \mathcal{B} v \rangle + v^T \mathcal{B}^* q_c = 0, \\ W_\tau = \mathcal{L}_\kappa W + g \implies \|W(\tau)\|_{CR} \le \|W(0)\|_{CR} + \int_0^\tau \|g(s)\|_{CR} \, \mathrm{d}s.\end{gathered} \tag{5.4}$$

For a known strong curve, the derivative of $\sqrt{\|W\|_{CR}^2 + \delta^2}$ is at most $\|g\|_{CR}$. Integration and $\delta \downarrow 0$ prove the last line at zeros of W as well. These are the weighted estimates of the published proof [2, Section 4.2]; receiver loading remains inside the norm-preserving linear part.

Empty external preparation and conservation give $|x_\kappa| \le R$. The derivative $J(x)h = K(h_a \times b + a \times h_b)$ implies

$$|f(x)| \le \frac{k}{2}|x|^2, \quad \|J(x)\| \le k|x|, \quad |f(x) - f(\widetilde{x})| \le kR|x - \widetilde{x}| \quad (|x|, |\widetilde{x}| \le R). \tag{5.5}$$

The last bound integrates J on the line segment in the convex ball; it does not require preservation of a coherent phase sector. Applying (5.4) with forcing $(\mathcal{B}f(x_\kappa), 0, 0)$ and then integrating the native equation gives, for every finite $\tau \ge 0$,

$$\begin{gathered}\|V_\kappa(\tau)\|_{CR} \le \sqrt{m} F_{\max} \tau, \qquad |\widehat{y}(\tau)| \le m F_{\max} \tau, \\ |x_\kappa(\tau) - x_0| \le \frac{\eta_N^2 k R m F_{\max}}{4\alpha_N \kappa^2} \tau^2, \qquad |f(x_\kappa(\tau)) - f_0| \le \frac{A\tau^2}{\kappa^2}.\end{gathered} \tag{5.6}$$

The coefficient one-fourth contains the feedback factor one-half and the integral of time. These exact-solution bounds are inherited from the published argument [2, Sections 4.2–4.3] and are not restricted to its witness horizon.

5.2 The longer comparison window

Freeze the core and omit scaled propagation only in the comparison:

$$\begin{aligned} \widehat{p}^{(0)}(\tau) &= \frac{\sin(\omega\tau)}{\omega}\mathcal{B}f_0, & \widehat{r}^{(0)}(\tau) &= 0, \\ \widehat{z}^{(0)}(\tau) &= f_0(1-\cos(\omega\tau)), & \widehat{y}^{(0)}(\tau) &= \frac{m}{\omega}f_0\sin(\omega\tau). \end{aligned} \tag{5.7}$$

All its external coordinates vanish initially. It solves its comparison equations because $\omega^2 = m/\beta_R$ [2, Section 5]. It is not the exact finite-speed solution.

Let $E_\kappa = V_\kappa - V_0$. The exact empty-preparation solution has the strong regularity (2.5), preserved by finite scaling. The explicit comparison carrier is $C^1_\tau H^1_s$. Their difference therefore already has carrier regularity $C_\tau H^1_s \cap C^1_\tau L^2_s$ and a C^1 receiver coordinate, before estimating its residual. Subtraction gives

$$\begin{aligned} (E_\kappa)_\tau = \mathcal{L}_\kappa E_\kappa &+ (\mathcal{B}(f(x_\kappa) - f_0), 0, 0) \\ &+ \left(0, -\frac{c}{\kappa\omega}\mathcal{B}' f_0 \sin(\omega\tau), 0\right), \qquad E_\kappa(0) = 0. \end{aligned} \tag{5.8}$$

The propagation residual is negative in the second carrier coordinate; the first is zero because $\widehat{r}^{(0)} = 0$. It lies in L^2 through the first weak derivative of the tents. It need not lie in H^1. We neither differentiate it into H^1 nor infer strong regularity from it. No second tent derivative or operator-norm limit of an unbounded generator is used.

The forced estimate and (5.6) imply, for $0 \le \tau \le S$,

$$\begin{aligned} \|E_\kappa(\tau)\|_{CR} &\le \frac{c\sqrt{d}\,|f_0|}{\kappa\omega}\int_0^\tau |\sin(\omega s)|\,\mathrm{d}s + \frac{\sqrt{m}\,A\tau^3}{3\kappa^2} \\ &\le \frac{4c\sqrt{d}\,|f_0|}{\kappa\omega^2} + \frac{\sqrt{m}\,AS^3}{3\kappa^2} = \varepsilon_\kappa. \end{aligned} \tag{5.9}$$

Each half of $[0, 2\pi/\omega]$ contributes $2/\omega$ to the absolute sine integral. Both integrated bounds are nondecreasing with the upper endpoint. Thus (5.9) is uniform over the whole window. Global strong existence and the all-finite-τ drift estimates justify extending the argument; the proof does not change the horizon in the published native-credit theorem.

5.3 Weighted projections and positive denominators

By $\mathcal{B}^*\mathcal{B} = m\mathrm{Id}_2$ and $m/\omega^2 = \beta_R$,

$$\|V_0(\tau)\|^2_{CR} = \Lambda^2[\sin^2(\omega\tau) + (1-\cos(\omega\tau))^2] = 4\Lambda^2\sin^2(\omega\tau/2). \tag{5.10}$$

Its norm is at most 2Λ, equals 2Λ at π/ω, and the entire comparison state is zero at S. Triangle and reverse triangle inequalities, followed by the stock conversion (5.2), prove (4.5) whenever $\varepsilon_\kappa < 2\Lambda$. Positivity is established before the lower bound is squared.

The carrier projection is contractive in the L^2 pair norm, and the receiver projection is contractive in $\sqrt{\beta_R}$ times its Euclidean norm. At π/ω, the comparison carrier is zero and the comparison receiver is $2f_0$, of weighted norm 2Λ. The same error estimate therefore proves (4.6). The small carrier bound uses its intermediate zero; the endpoint estimate uses the whole-state zero at S. Receiver positivity is separate from total-stock positivity.

The definitions (4.7) now have positive denominators. Conservation gives the recovery identity there. Since $1 - \mathcal{S}_\kappa = H_C(t_\pi)/E_{\mathrm{out}}(t_\pi)$ and $H_R(t_{2\pi}) \le E_{\mathrm{out}}(t_{2\pi})$, division of the appropriate upper and lower bounds proves the first line of (4.8). It holds for $\varepsilon_\kappa < 2\Lambda$, with a strictly

positive common fraction lower bound when $\varepsilon_\kappa < \Lambda$. Subtracting the endpoint upper bound from the intermediate lower bound gives

$$\begin{aligned} H_N(t_{2\pi}) - H_N(t_\pi) &= E_{\mathrm{out}}(t_\pi) - E_{\mathrm{out}}(t_{2\pi}) \\ &\geq a_\kappa[(2\Lambda - \varepsilon_\kappa)^2 - \varepsilon_\kappa^2] = \frac{2\eta_N^2}{\kappa^2}\Lambda(\Lambda - \varepsilon_\kappa). \end{aligned} \tag{5.11}$$

The increment is at most $E_{\mathrm{out}}(t_\pi) \leq M_\kappa$. No monotonicity throughout the interval is used.

5.4 A separate receiver-product estimate

At $\tau_{\mathrm{q}} = \pi/(2\omega)$ and $\tau_{\mathrm{b}} = 3\pi/(2\omega)$ the comparison receiver is f_0 and its output is respectively $+(m/\omega)f_0$ and $-(m/\omega)f_0$. Put $\delta z = \widehat{z} - \widehat{z}^{(0)}$ and $\delta y = \widehat{y} - \widehat{y}^{(0)}$. Contractive projection and bounded $\mathcal{B}^*$ give $|\delta z| \leq \varepsilon_\kappa/\sqrt{\beta_R}$ and $|\delta y| \leq \sqrt{m}\,\varepsilon_\kappa$. The complete product difference is

$$\widehat{z}^T\widehat{y} - (\widehat{z}^{(0)})^T\widehat{y}^{(0)} = (\widehat{z}^{(0)})^T\delta y + \delta z^T\widehat{y}^{(0)} + \delta z^T\delta y. \tag{5.12}$$

The first two terms each have bound $\sqrt{m}\,|f_0|\varepsilon_\kappa$, since $m/(\omega\sqrt{\beta_R}) = \sqrt{m}$. The third has bound $\omega\varepsilon_\kappa^2$. Hence, writing $Q_R = \widehat{z}^T\widehat{y}$,

$$\begin{aligned} |Q_R(\tau_{\mathrm{b}}) + D| &\leq \omega(2\Lambda\varepsilon_\kappa + \varepsilon_\kappa^2), \\ |Q_R(\tau_{\mathrm{q}}) - D| &\leq \omega(2\Lambda\varepsilon_\kappa + \varepsilon_\kappa^2). \end{aligned} \tag{5.13}$$

Both vector errors and their product are included. The power factor in (5.2) contains one κ from the definition of P_{CR} and two factors η_N/κ from the states.

5.5 Threshold, native import and chronology

For $\kappa \geq \kappa_{\mathrm{cert}}(q)$, the two allocations are

$$\frac{B_1}{\kappa} \leq \frac{q\Lambda}{2}, \qquad \frac{B_2}{\kappa^2} \leq \frac{q\Lambda}{2}, \qquad \varepsilon_\kappa \leq q\Lambda < \Lambda, \qquad 2q + q^2 \leq \frac{11}{25} < \frac{1}{2}. \tag{5.14}$$

Thus $r_\kappa \leq [q/(2-q)]^2$ and the second line of (4.8) follows. The error in (5.13) is at most $D(2q + q^2)$, leaving a sign margin at least $14D/25 \geq D/2$. This proves the receiver witnesses (4.10).

The native witness (4.11) follows from the same-trajectory paper's Theorem 3.1 [2], with exactly its coherent launch, constants (4.2) and $T/\kappa = t_{\mathrm{b}}$. That theorem is not reproved or mistaken for the receiver-power estimate.

For each fixed finite κ, the published empty-launch proposition [1, Proposition 4.1] gives initial $P_{NC} > 0$, $H_R > 0$ and $P_{CR} > 0$. Shorten their common interval into $(0, t_{\mathrm{q}}/2)$. The separate finite estimate proves charging at t_{q}; no nonuniform fixed-κ short-time expansion is evaluated at a κ-dependent landmark.

Continuity of z and of $\mathcal{B}^*p$ makes P_{CR} continuous. Its strict negative value at t_{b} permits

$$\begin{aligned} 0 < \rho_\kappa < \frac{\pi}{4\omega\kappa}, \qquad & [t_{\mathrm{b}} - \rho_\kappa, t_{\mathrm{b}} + \rho_\kappa] \subset (t_\pi, t_{2\pi}), \\ P_{CR}(t) \leq -\frac{\eta_N^2 D}{4\kappa} \quad & (t \in [t_{\mathrm{b}} - \rho_\kappa, t_{\mathrm{b}} + \rho_\kappa]). \end{aligned} \tag{5.15}$$

Integration gives strict receiver-storage decrease on this interval. By continuity of P_{NC} and its margin, the radius can be further shortened to a common interval with the same negative native-power bound. No explicit or uniform positive lower bound for its width is claimed. Uniqueness places initial supply, receiving, later backflow and recovery on one solution with unchanged coefficients. This completes Theorem 4.1.

6 Absolute amount, relative recovery and structural ceilings

6.1 The original accessibility normalization

For the specified coherent launch, the old invariant ceiling has the fixed positive value

$$\begin{aligned} C_{\mathrm{inv}} &= \frac{\alpha_N}{2}(3-\sqrt{9-8\sin^2\phi}) = C_Z > 0, \qquad \mathcal{A}_\kappa = \frac{M_\kappa}{C_{\mathrm{inv}}}, \\ \frac{a_\kappa(2\Lambda-\varepsilon_\kappa)^2}{C_{\mathrm{inv}}} &\le \mathcal{A}_\kappa \le \min\left\{1, \frac{a_\kappa(2\Lambda+\varepsilon_\kappa)^2}{C_{\mathrm{inv}}}\right\}. \end{aligned} \tag{6.1}$$

The displayed finite bounds require $\varepsilon_\kappa < 2\Lambda$ and in particular hold in the certified regime. Positivity follows from $0 < \phi < \pi/12$. This denominator is the published upper ceiling, not a proved attainable maximum.

The two accepted results combine directly:

$$M_\kappa \le C_{\mathrm{joint}} \le C_Z = C_{\mathrm{inv}}. \tag{6.2}$$

Indeed the joint all-time bound holds at every time in the compact interval defining M_κ; its minimum-of-ceilings form makes it no larger than C_Z. Equation (6.1) retains its original denominator. No new normalized statistic is needed.

6.2 Finite errors before asymptotics

Put $L_{\mathrm{err}} = B_1 + B_2$, so $\varepsilon_\kappa \le L_{\mathrm{err}}/\kappa$ for $\kappa \ge 1$. In the certified regime the stock sandwich, projections and positive denominators yield

$$\begin{aligned} \left| M_\kappa - \frac{2\eta_N^2\Lambda^2}{\kappa^2} \right| &\le \frac{\eta_N^2}{2\kappa^2}(4\Lambda\varepsilon_\kappa + \varepsilon_\kappa^2) \\ &\le \frac{2\eta_N^2\Lambda L_{\mathrm{err}}}{\kappa^3} + \frac{\eta_N^2 L_{\mathrm{err}}^2}{2\kappa^4}, \\ 0 \le E_{\mathrm{out}}(t_{2\pi}) &\le \frac{\eta_N^2 L_{\mathrm{err}}^2}{2\kappa^4}. \end{aligned} \tag{6.3}$$

The lower and upper deviations of the squared stock sandwich are bounded by the same $4\Lambda\varepsilon_\kappa+\varepsilon_\kappa^2$. Since $2\Lambda - \varepsilon_\kappa \ge (2-q)\Lambda > 0$,

$$\begin{aligned} 0 \le 1-\mathcal{S}_\kappa,\ 1-\mathcal{R}_\kappa,\ \frac{H_R(t_{2\pi})}{H_R(t_\pi)} &\le \frac{L_{\mathrm{err}}^2}{(2-q)^2\Lambda^2\kappa^2}, \\ 0 < \mathcal{A}_\kappa &\le \min\left\{1, \frac{\eta_N^2(2+q)^2\Lambda^2}{2C_{\mathrm{inv}}\kappa^2}\right\}. \end{aligned} \tag{6.4}$$

All constants are fixed in the following limits:

$$\begin{aligned} M_\kappa &= \frac{2\eta_N^2\beta_R|f_0|^2}{\kappa^2} + O(\kappa^{-3}), \qquad E_{\mathrm{out}}(t_{2\pi}) = O(\kappa^{-4}), \\ \mathcal{S}_\kappa, \mathcal{R}_\kappa &\longrightarrow 1, \qquad 1-\mathcal{S}_\kappa,\ 1-\mathcal{R}_\kappa = O(\kappa^{-2}), \qquad \mathcal{A}_\kappa \longrightarrow 0, \\ 2\eta_N^2\beta_R|f_0|^2 &= \frac{2\eta_N^2\beta_R G^2\sin^2\phi}{7}(6+8\sin^2\delta_g). \end{aligned} \tag{6.5}$$

The leading transferred-stock coefficient is quadratic in the native coupling and launch-drive magnitude and linear in the receiver storage scale. The receiver-coupling dependence is inverse

square with every other parameter fixed. This is an asymptotic scale, not a monotonicity claim at every finite coupling or a uniform limit as $\phi \to 0$.

The 80/81 guarantee concerns recovery of actual intermediate stock. Multiplying it by C_{joint} would not produce an established recovered amount. At the strict witness yaw the stronger ceiling remains positive and independent of κ, so it can coexist with this small transfer and high recovered fraction. The entire physical observation window shrinks relative to the fixed circuit time:

$$\frac{t_{2\pi}}{L_c/c} = \frac{cS}{\kappa L_c} \longrightarrow 0. \tag{6.6}$$

Both couplings are collocated on the finite profiles. The quantitative certificate therefore describes fast reversible exchange, not a wave completing a carrier circuit before recovery.

7 Interpretation and conclusion

The fixed rank-two candidate has a second native invariant that sharpens a necessary all-time net-extraction ceiling without a large-coupling assumption. Its joint algebraic floor is exact in the full common invariant level, yet the active zero-level example shows that an algebraic minimizer need not be reached at finite time. Actual attainment of C_{joint} and infinite-time approach remain unresolved.

The separate quantitative theorem concerns the specified coherent family with initially empty carrier and receiver and an explicit sufficient finite-coupling regime. It establishes actual temporary receiver concentration, charging, backflow and net native recovery. A large recovered fraction accompanies a small stock in a shrinking window. Under empty preparation H_N never exceeds its initial value. Net capacity differs from gross repeated positive work, which may count storage returned and supplied again. No exact nonlinear period, exact maximizing time, monotonic recovery over the whole window, permanent retention, disposal or full-time optimized extraction is proved.

The author's intended geometry starts from a central meeting region with opposed cones, outward spreading and a curved around–inward return, stationary supports and possible transfer elsewhere. The carrier circle remains distinct from the intended tight toroid, the close original enclosing sphere from the extended-line sphere, and finite host-notch N regions from the central broken G boundaries [1, Section 8]. The kernel direction e has no assigned physical support, opening, field axis or receiving location. The tentative 15–20%, possibly more, is geometric intuition, not a fitted target or a consequence of the fractions proved here.

Useful accessible capacity, native stepping/cadence, intended route correspondence and six-opening transport remain the next research directions. Linked gravity–magnetism, outer geometry and justified SRG integration remain in the continuing programme. This manuscript makes no new geometric choice or scientific generalization.

Contributions and evidence

Hilmir Frímann Halldórsson (pzychozen) supplies research authorship and direction. GPT formulated and internally reviewed the investigations; Codex independently investigated them and prepared this manuscript [3, 4, 5, 6]. The accepted evidence contains 18 AT and 12 PC grouped algebra checks, each independently replayed. Replays and document checks add no mathematical identities; written proofs and cited foundations carry the claims. No scientific replay or simulation was performed for this manuscript. The underlying mathematics and submitted manuscript are internally accepted; the author has authorized this publication edition. No external peer review or physical validation is claimed.

A Dimensions and notation

Use the predecessor's abstract units $\mathrm{U}, \mathrm{L}, \mathrm{T}$ and native amplitude unit $\mathrm{A}_{\mathrm{amp}}$ [1, Section 1]. The normalized coherent launch uses unit native amplitudes. The drift constant A differs from $\mathrm{A}_{\mathrm{amp}}$ and A_{reg}.

Quantity	Dimension
x, R; $I, h_e, \mathcal{J}_K, Z_2$	$\mathrm{A}_{\mathrm{amp}}$; $\mathrm{A}_{\mathrm{amp}}^2$
K, k; J	$\mathrm{UL}/(\mathrm{TA}_{\mathrm{amp}}^2)$; $\mathrm{UL}/(\mathrm{TA}_{\mathrm{amp}})$
p, r_c, y; $f, z, F_{\max}$	U; UL/T
m, d, c; ω	$\mathrm{L}^{-1}, \mathrm{L}^{-3}, \mathrm{L/T}$; T^{-1}
α_N; β_R	$\mathrm{U}^2\mathrm{L}/\mathrm{A}_{\mathrm{amp}}^2$; T^2/L
A; $\lVert V \rVert_{CR}, \Lambda, B_1, B_2, \varepsilon_\kappa, L_{\mathrm{err}}$	$\mathrm{UL/T}^3$; $\mathrm{UL}^{1/2}$
$D, C_1, C_2, P_{NC}, P_{CR}$	$\mathrm{U}^2\mathrm{L/T}$
$H_N, H_C, H_R, M_\kappa, C_Z, C_{\mathrm{joint}}, C_{\mathrm{inv}}$	$\mathrm{U}^2\mathrm{L}$
$\tau, S, T, t_{\mathrm{q}}, t_\pi, t_{\mathrm{b}}, t_{2\pi}, \rho_\kappa$	T
$e, \eta_N, \kappa, q, a_\kappa, \kappa_{\mathrm{crit}}, \kappa_{\mathrm{cert}}$	1
$r_\kappa, \mathcal{S}_\kappa, \mathcal{R}_\kappa, \mathcal{A}_\kappa$; retention ratio	1

Table 1: Inherited bookkeeping and dimensions of the combined certificate.

In particular B_1/Λ, B_2/Λ, C_1/D and C_2/D are dimensionless. The structural projection account has storage units and adds no constitutive energy. These are commensurability checks, not physical calibration. The notation L_{err} is only an alias for the accepted error-constant sum $L = B_1 + B_2$, used to distinguish it from carrier length L_c.

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